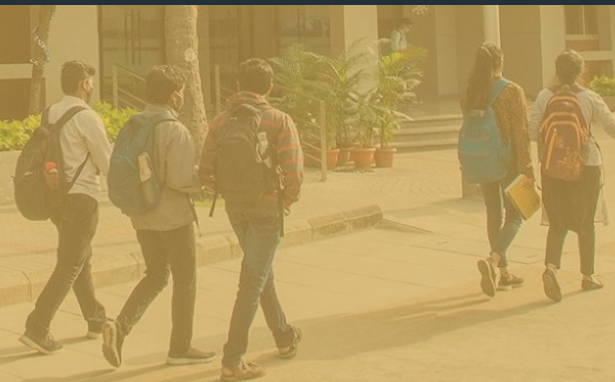


NEW-AGE GLOBAL UNIVERSITY FOR LIBERAL EDUCATION



General Information

CS1003 – Programming in C

Credits – 4, L T P – 3 0 2

Internal Assessment – 70 Marks

Semester End Examination – 30 Marks

Minimum Marks to Pass

Internal

Semester End

28

12

Course Outcomes

Course Outcomes: After completing the course, the students will be able to:

a.	Understand and analyse the concepts of the number system, computer memory, and the fundamental concepts of the C programming language.
b.	Demonstrate proficiency in using fundamental data types, operators, and console input/output functions in C
c.	Design and implement C programs using decision-making constructs, loops, arrays, strings, functions, and pointers to solve mathematical and logical problems.
d.	Analyse and evaluate the programming concepts based on user-defined data types and file handling in C language.

Module – 1: Computer Fundamentals in C Language	No. of hours
	3
Fundamentals of Computers: C Language: Structure of C Program, Memory Layout of C Programs, Phases of developing a running computer program in C, how does a C Program Execute: Pre-processor, Compiler, Linker, Loader, Library Functions, Various components of a computer that programs require for execution.	

Module – 2: Data Concepts and Operators	No. of hours
	6
Data concepts in C: Constants, Variables - Variables as named memory locations, Rules for Constructing Variables, understanding different Data Types - range, size, and appropriate usage, C Keywords, format specifiers, Uses and Common Errors associated with them, Instruction, Pre-processor Directive, Macros. Operator in C: Arithmetic Operators - Hierarchy of operations, Precedence, Relational Operators, Logical Operators, Conditional Operators, Bit-wise Operators, Assignment Operators. Type Conversion and Type Casting.	

Syllabus Contd...

Module – 3: Decision Control, Loops, and Functions	No. of hours
	14
Decision Control in C: if Statement, else-if Statement, nested else -if Statement - Use case, example for nested if-else, switch. Loop statements: for, while, do-while, break (from loop), continue, exit - exit status to represent the exit code of the program, program clean ups done by exit function. Introduction to Functions: Function Declaration and Definition. Function Prototype. Calling a Function, Return Value. Parameters and Arguments. Local, Global, and Static Variables. Pass by value and reference. Understanding the use of pointers. Recursion: Basic concept of recursion, Study of basic problems using recursion: factorial, GCD, Fibonacci series.	

Syllabus Contd...

Module – 4: Arrays and Strings	No. of hours
	12
Arrays: Declaration and Initialization, Indexing, Array Size and Bounds. One-dimensional, Two-dimensional and Multi-dimensional arrays. passing array elements to functions, passing entire array to functions, Basic array operations- traversal, insertion, deletion, updation. Array manipulation- reversing, concatenation, splitting, sorting. Array library functions. String: Strings as array of characters, pointers and strings, string operations. String library functions.	

Module – 5: Structures and File operations	No. of hours
	10
Structures: Introduction to Structures and Unions, Structure Definition and Declaration, Accessing Structure Members. Array of Structures. Passing Structures to Functions. Basic file operations: Creating a new file, opening an existing file, reading from file, writing to a file, moving to a specific location in a file, Closing a file.	

Laboratory Component

1. Write a program using function-like macros to add two numbers.
2. Write a program to swap two numbers with and without using a third variable.
3. Write a program to create a menu-driven basic arithmetic calculator.
4. Write a program to perform the below tasks using loops
 - a. Print a Fibonacci series.
 - b. Compute GCD of two numbers.
5. Write a program to perform the below tasks using functions
 - a. Print a Fibonacci series.
 - b. Compute GCD of two numbers.

Laboratory Component Contd...

6. Write a program to perform the below tasks using recursion
 - a. Print a Fibonacci series.
 - b. Compute GCD of two numbers.
7. Write a menu driven program to demonstrate the following array operations using functions: Insert, Delete, Update element at index i.
8. Write a program to implement different string operations using functions.
9. Write a program using structures to read, write and compute average-marks of n students. Take student name, USN, and marks in 5 subjects as input. Pass structure by reference to a function to update marks in one of the five subjects as given by user for a particular student. Finally display the new average for that student.
10. Write a program to demonstrate different file operations in C.

CIE Components

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Component 1	Self-learning module completion for 5 marks. Specified modules in the CodeTantra must be completed within the given timeframe. Faculty will verify module completion through the CodeTantra faculty interface (<i>Refer Table-1</i>)
	Component 2	Test will be conducted for 15 marks and will be evaluated according to the Scheme of Evaluation prepared by the team of course faculty.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the team of course faculty.

CIE Components Contd...

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Component 1	Self-learning module completion for 5 marks. Specified modules in the CodeTantra must be completed within the given timeframe. Faculty will verify module completion through the CodeTantra faculty interface (<i>Refer Table-1</i>)
	Component 2	Continuous Lab assessment + Viva for 5 marks (<i>Refer Table-2</i>) (10 lab programs 10 marks each, 100 marks reduced to 5 marks)
	Component 3	Internal lab practical exam for 15 marks (<i>Refer Table-2</i>) (3 lab programs 10 marks each, 30 marks reduced to 15 marks)

CIE Components Contd...

Table-1:

CIE 1 and CIE 3 Graded Component - 1 (5 Marks):

Assess students' ability to complete assigned self-learning modules on the CodeTantra platform within the given timeframe.

Criteria	Fully Completed (5 Marks)	Mostly Completed (3-4 Marks)	Partially Completed (1-2 Marks)	Incomplete (0 Marks)
Completion of self-learning modules	Successfully completes all modules on the CodeTantra platform before the deadline.	Completes more than 50% of the modules.	Completes less than 50% of the modules.	Fails to complete any module within the given time.

Lab assessment + Viva (10 Marks per program)				
Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0 Marks)
Understanding of problem statement (2 Marks)	Observations	The student exhibits a thorough understanding of the requirements for the problem.	The student has a sufficient understanding of the requirements for the problem.	The student does not have a clear understanding of the requirements for the problem.
Execution (2 Marks)	Observations	The Student demonstrates the execution of the program with optimized code and shows the output with appropriate validations and test cases handled.	The Student demonstrates the execution of the program without optimization of the code and shows the output.	The Student has not executed the program.
Results and Documentation (2Marks)	Observations	Documentation with appropriate comments and output cases are noted in data sheets.	Documentation with few comments and few output cases are noted in data sheets.	Documentation with no comments and no output cases are noted in data sheets.
Viva (4 Marks)	Observations	The student is eager to answer all questions and provides meaningful answers to the questions asked (4– 3 M)	The student tries to answer the questions partially and answers with some cues and help. (2-1 M)	The student does not answer the questions even with cues and help.

Text Books:

1. Let Us C: Authentic guide to C programming language by Yashavant Kanetkar. BPB Publications; Nineteenth edition (15 December 2022); BPB Publications, Ansari Road, Dariya Ganj. ISBN-10: 9355512767. ISBN-13: 978-9355512765.
2. C Programming Absolute Beginner's Guide by Greg Perry and Dean Miller. Que Publishing; 3rd edition (22 August 2013). ISBN-10: 0789751984. ISBN-13: 978-0789751980.
3. Computer Programming in C by Rajaraman V. Prentice Hall India Learning Private Limited (1 January 1994). ISBN-10: 9788120308596. ISBN-13: 978-8120308596

Reference books:

1. How To Solve It By Computer by R G Dromey. ISBN-10: 8120303881. ISBN-13: 978-8120303881
2. Problem Solving and Programming Concepts by Sprankle, Pearson Education India; 9th edition (1 January 2013). ISBN-10: 9789332518841. ISBN-13: 978-9332518841
3. The C Programming Language by Brian W. Kernighan and Dennis Ritchie. Pearson Education India; 2nd edition (1 January 2015). ISBN-10: 9332549443. ISBN-13: 978-9332549449`

Grades and Grade Points

Undergraduate Degree	Letter Grade	Grade Point
90 – 100	O (Outstanding)	10
80 – 89	A+ (Excellent)	9
70 – 79	A (Very Good)	8
60 – 69	B+ (Good)	7
50 – 59	B (Above Average)	6
45 – 49	C (Average)	5
40 – 44	P (Pass)	4
< 40	F (Fail)	0

$$SGPA = \frac{\sum_i C_i GP_i}{\sum_i C_i}$$

C_i = Credits assigned to the course.

GP_i = Points corresponding to the grade obtained in the course.

Students Attendance Policy

1. Minimum Attendance Requirement

- Minimum **80% attendance per course** is mandatory for SEE (Semester End Examination) eligibility.
- Attendance is counted only for **physical presence in class**.
- **20% margin** is permitted for:
 - Illness, bereavement, family ceremonies, personal commitments (e.g., marriage), etc.
 - Students must **inform mentors via email** at the earliest regarding any absence.
- Students with **<80% attendance** will be placed on the **SA list** and must repeat the course during the summer.

Students Attendance Policy Contd..

2. Special Considerations for Attendance

Case 1: Prolonged Medical Conditions

- Inform mentor via parent's registered email ID with medical proof.
- Confirm receipt of the email with mentor via phone call.
- Send periodic health updates to mentor.
- Submit a complete medical report upon resuming classes.
- Parent/guardian meetings with mentor & authorities help in faster decisions.

Case 2: Recognized Event Participation (intercollegiate or university-recognized events)

- Must have $\geq 80\%$ attendance when requesting permission.
- Obtain prior approval from mentor & Program Director.
- Submit valid participation proof post-event.

Students Attendance Policy Contd..

3. Internal Attendance Committee Review

- Attendance consideration request along with all required documents submitted by students will be reviewed weekly.
- Committee's decision is final and will be communicated to students and parents.
- Committee holds sole discretion on leave approval or rejection.
- Attendance consideration request along with documents must be sent to the respective mentor by the student.

Students Attendance Policy Contd..

4. CIE (Continuous Internal Evaluation) Policy

- Total CIE marks: 70 (through CIE1, CIE2, CIE3).
- No re-exams for missed CIEs during the semester.
- No improvement tests will be conducted.
- Students must aim to score at least 40% in regular CIEs.

5. Re-test eligibility (end of semester):

- Students with genuine reasons only
- Minimum 80% attendance.
- Attendance in at least one regular CIE.
- Written permission from Dean required.

Thank You